

Report di laboratorio

Familiarizzazione con Linux, Shell e comandi da terminale tramite GameShell

Studente: Vincenzo Lembo

Corso: Cybersecurity & Ethical Hacking

Esercitazione: W8D4 - Progetto finale

Argomento: GameShell su Kali Linux

Ambiente: Kali Linux – UTM

1. Obiettivo dell'esercitazione

L'obiettivo dell'esercitazione è stato quello di prendere familiarità con il sistema operativo **Kali Linux** e con l'utilizzo della **Shell**, imparando alcuni dei principali comandi Linux attraverso GameShell.

GameShell è un gioco strutturato in diverse missioni che richiedono l'utilizzo progressivo dei comandi del terminale. Per ogni missione è possibile visualizzare l'obiettivo tramite `gsh goal` e verificarne il completamento tramite `gsh check`.

Durante l'esercitazione ho completato le missioni previste fino alla **missione 15**, utilizzando i diversi comandi introdotti dal gioco.

2. Installazione di GameShell

Prima di procedere con l'installazione ho aggiornato l'elenco dei pacchetti:

```
sudo apt update
```

Successivamente ho installato i pacchetti necessari:

```
sudo apt install gettext man-db procps psmisc nano tree bsdmainutils x11-apps wget
```

Ho quindi scaricato GameShell:

```
wget https://github.com/phyver/GameShell/releases/download/latest/gameshell.sh
```

Infine ho avviato il gioco:

```
bash gameshell.sh
```

Questa procedura segue le indicazioni fornite nella traccia dell'esercitazione.



3. Svolgimento delle missioni

Missione 1 – Navigazione tra le directory

```
| Mission goal  
|-----  
|  
| Go to the top of the main tower of the castle.  
|  
| Useful commands  
|-----  
|  
| cd LOCATION  
| Move to the given location.  
| Remark: `cd` is an abbreviation for "change directory".  
|  
| pwd  
| Show the path to your current location.  
| Remark: `pwd` is an abbreviation for "print working directory".  
|  
| ls  
| Show a list of locations that are currently accessible.  
| Remark: `ls` is an abbreviation of "list".  
|  
| gsh check  
| Check if the mission objective has been achieved.  
|  
| gsh reset  
| Restart the mission from the beginning.  
|  
| Remarks  
|-----  
|  
| UPPERCASE words appearing in commands are meta-variables: you need to replace them by  
| appropriate (string) values.  
|  
| Most filesystems treat uppercase and lowercase characters differently. Make sure you use the  
| correct path.  
|
```

Nella prima missione dovevo raggiungere la cima della torre principale del castello.

Per orientarmi nel filesystem ho utilizzato:

pwd

ls

`pwd` permette di conoscere la directory corrente, mentre `ls` mostra i file e le directory presenti.

Sono quindi entrato nelle varie directory tramite `cd`:

`cd Castle`

`cd Main_tower`

`cd First_floor`

`cd Second_floor`

`cd Top_of_the_tower`

Infine ho verificato il completamento:

`gsh check`

Questo procedimento è anche quello illustrato nella traccia dell'esercitazione.

```
[mission 1] $ pwd
/home/kali/gameshell/World

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Castle Forest Garden Mountain Stall

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Castle

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Cellar Great_hall Main_building Main_tower Observatory

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Main_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
First_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd First_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Second_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Second_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ █
```

Missione 2 – Navigazione nel filesystem

Nella seconda missione ho continuato a utilizzare i comandi di navigazione imparati precedentemente, muovendomi tra le directory del castello e approfondendo l'utilizzo di `cd`, `ls` e dei percorsi Linux.

```
[use 'gsh help' to get a list of available commands]
[mission 2] $ cd /home/kali/gameshell/World

[use 'gsh help' to get a list of available commands]
[mission 2] $ pwd
/home/kali/gameshell/World

[use 'gsh help' to get a list of available commands]
[mission 2] $ ls
Castle Forest Garden Mountain Stall

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd Castle

[use 'gsh help' to get a list of available commands]
[mission 2] $ ls
Cellar Great_hall Main_building Main_tower Observatory

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd Cellar

[use 'gsh help' to get a list of available commands]
[mission 2] $ gsh check

Congratulations, mission 2 has been successfully completed!
progress was saved in /home/kali/gameshell-save.sh ]
```

Missione 3 – Utilizzo dei percorsi

```
[mission 3] $ gsh goal

Mission goal
Go back to the starting location and then go to the throne room using only two commands.

Remark
You may experiment with as many commands as you want, but
to validate the mission the following conditions need to be met:
- the second to last command takes you to the starting point,
- the last command takes you directly to the throne room.

Useful commands
cd
Move back to GameShell's starting location.
cd LOCATION1/LOCATION2/LOCATION3
Make several moves in one command.

Remark
UPPERCASE words appearing in commands are meta-variables: you need to replace them by
appropriate (string) values.
```

L'obiettivo era tornare alla posizione iniziale e raggiungere la sala del trono utilizzando solamente due comandi.

Ho utilizzato:

cd

cd Castle/Main_building/Throne_room

Questa missione mi ha permesso di capire che con **cd** è possibile attraversare più directory contemporaneamente indicando il percorso completo.

```

[mission 3] $ cd

[use 'gsh help' to get a list of available commands]
[mission 3] $ cd Castle/Main_building/Throne_room

[use 'gsh help' to get a list of available commands]
[mission 3] $ gsh check

Congratulations, mission 3 has been successfully completed!

```

Missione 4 – Creazione delle directory

```

~/Castle/Main_building/Throne_room
[mission 4] $ gsh goal

/8\ \_..
( \_/_/  || Mission goal
||
|| Build a "Hut" in the forest, and then build a "Chest" in the hut.
||
|| Useful commands
||
|| mkdir DIRECTORY
|| Create a new directory inside the current directory.
|| Remark: ``mkdir`` is an abbreviation for "make directory".
||
/8\ \_..
( \_/_/  ||
||
|| \_/_/
||

```

In questa missione dovevo costruire una **Hut** nella foresta e successivamente un **Chest** al suo interno.

Ho utilizzato:

cd

ls

cd Forest

mkdir Hut

cd Hut

mkdir Chest

Il comando **mkdir** (*make directory*) permette di creare una nuova directory.

```
~/Castle/Main_building/Throne_room
[mission 4] $ cd

~
[mission 4] $ ls
Castle Forest Garden Mountain Stall

~
[mission 4] $ cd Forest
~/Forest
[mission 4] $ mkdir Hut
~/Forest
[mission 4] $ cd Hut
~/Forest/Hut
[mission 4] $ mkdir Chest
~/Forest/Hut
[mission 4] $ gsh check

Congratulations, mission 4 has been successfully completed!
[ progress was saved in /home/kali/gameshell-save.sh ]
```

Missione 5 – Eliminazione dei file

```
(\
  Mission goal
  _____
  Go back to the cellar and get rid of all the spiders. Leave the bats alone: they appear on
  the castle's coat of arms and are said to confer luck.

  Useful commands
  _____
  rm FILE1 FILE2 ... FILEn
  Delete the files (permanently).
  Remark: "rm" is an abbreviation for "remove".
  (*)
  )
```

Nella cantina del castello erano presenti alcuni ragni e pipistrelli. L'obiettivo era eliminare solamente i ragni.

Dopo aver controllato i file con `ls`, ho utilizzato:

```
rm spider_1 spider_2 spider_3
```

Il comando `rm` (*remove*) permette di eliminare uno o più file.

```
~/Forest/Hut
[mission 5] $ cd

~
[mission 5] $ ls
Castle Forest Garden Mountain Stall

~
[mission 5] $ cd Castle
~/Castle
[mission 5] $ cd Cellar
~/Castle/Cellar
[mission 5] $ ls
barrel_of_apples bat_1 bat_2 spider_1 spider_2 spider_3
```

```
~/Castle/Cellar
[mission 5] $ rm spider_1 spider_2 spider_3

~/Castle/Cellar
[mission 5] $ ls
barrel_of_apples bat_1 bat_2

~/Castle/Cellar
[mission 5] $ gsh check

Congratulations, mission 5 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

|-----|
| Use the command |
| $ gsh help      |
| to get the list of "gsh" commands. |
|-----|
```

Missione 6 – Spostamento dei file

```
~/Castle/Cellar
[mission 6] $ gsh goal

Mission goal
=====
Collect all the coins that you can find in the garden in front of the castle, and put them
in your chest in your hut in the forest.

Useful commands
=====
mv FILE1 FILE2 ... FILEn DIRECTORY
Move the files to the directory.
Remark: ``mv`` is an abbreviation of "move".

~
The "~" symbol is an abbreviation for the initial directory.
Example: wherever you are, ``~/Tavern`` denotes the directory (or file) "Tavern" in the
initial directory.
```

Nel giardino erano presenti alcune monete che dovevano essere raccolte e spostate nel **Chest**.

Dopo aver individuato:

coin_1

coin_2

coin_3

le ho spostate utilizzando:

mv coin_1 coin_2 coin_3 ~/Forest/Hut/Chest

Il comando **mv** permette di spostare file da una directory a un'altra.

```
~/Castle/Cellar
[mission 6] $ ~/Garden
bash: /home/kali/gameshell/World/Garden: Is a directory

~/Castle/Cellar
[mission 6] $ cd

~/Castle/Cellar
[mission 6] $ ls
Castle Forest Garden Mountain Stall

~/Castle/Cellar
[mission 6] $ cd Garden

~/Garden
[mission 6] $ ls
coin_1 coin_2 coin_3 Flower_garden Maze Shed
```

```
~/Garden
[mission 6] $ mv coin_1 coin_2 coin_3 ~/Forest/Hut/Chest

~/Garden
[mission 6] $ gsh check

Congratulations, mission 6 has been successfully completed!
```

Missione 7 – File nascosti

```
(0) >>>=====
V)
( ) Mission goal
( ) Collect all the coins hidden in the garden in front of the castle, and put them in your
( ) chest (in your hut in the forest).
( )
( ) Secondary objective
( ) Learn how to use the "Tab" key to go faster.
( )
( ) Useful commands
( )
( ) ls -A
( ) List all the files of the current directory, including hidden files. (A file is "hidden"
( ) when its name starts with a dot.)
( )
( ) Tab
( ) The tabulation key "completes" the name of a file or directory once you have typed the
( ) beginning of its name. This only works
( ) if there is only one possible completion.
( )
( ) Tab-Tab
( ) Pressing tabulation twice successively shows a list of possible completions.
( )
(0) >>>=====
V)
```

In questa missione erano presenti altre monete nascoste nel Garden.

Con il normale **ls** non erano visibili, quindi ho utilizzato:

ls -A

In questo modo ho individuato i file nascosti, riconoscibili dal **.** iniziale, e li ho successivamente spostati nel Chest tramite **mv**.

```

~/Garden
[mission 7] $ ls -A
.17680_coin_2 .33731_coin_3 .37217_coin_1 Flower_garden Maze Shed

~/Garden
[mission 7] $ mv .17680_coin_2 .33731_coin_3 .37217_coin_1 ~/Forest/Hut/Chest

~/Garden
[mission 7] $ ls -A
Flower_garden Maze Shed

~/Garden
[mission 7] $ ls -A ~/Forest/Hut/Chest
.17680_coin_2 .33731_coin_3 .37217_coin_1 coin_1 coin_2 coin_3

~/Garden
[mission 7] $ gsh check
Error: unknown gsh command 'check'.
Use one of the following commands: check, goal, help, reset

~/Garden
[mission 7] $ gsh check
Congratulations, mission 7 has been successfully completed!

```

Missione 8 – Wildcard *

```

(0)=>======<=(0)
Mission goal
Get rid of all the spiders that are crawling in the cellar. Again, do not do not disturb the bats.

Shell patterns
*
The "*" character stands in for any sequence of characters
(including an empty sequence).
?
The "?" character stands in for any single character.
Those wildcards can be used to denote lists of existing files / directories in the current
working directory.
For example: if the current folder contains
file-1 Folder-1 file-14 potato
then
*      -> file-1 Folder-1 file-14 potato
*1     -> file-1 Folder-1
*0*    -> Folder-1 potato
x*     -> error, no matching file
*-?    -> file-1 Folder-1
*-??   -> file-14

```

La cantina conteneva numerosi ragni e alcuni pipistrelli. Per evitare di scrivere manualmente il nome di ogni ragno ho utilizzato un pattern:

```
rm *_spider_*
```

Il carattere ***** rappresenta una qualsiasi sequenza di caratteri. In questo modo ho potuto eliminare contemporaneamente tutti i file contenenti **_spider_**, lasciando intatti quelli relativi ai pipistrelli.

```

~/Garden
[mission 8] $ cd ~/Castle/Cellar

~/Castle/Cellar
[mission 8] $ ls
1026_spider_19  14525_bat_3      18427_spider_9  22584_spider_40  2666_spider_14  30768_spider_24  5273_spider_26
11082_spider_32 15617_spider_30 18548_bat_1     23145_spider_7   27616_spider_16 31462_spider_4   5574_spider_25
11387_spider_38 16449_spider_48 18699_spider_47 24272_spider_45  27710_spider_22 31501_spider_49  7306_spider_28
11813_spider_2  16654_spider_21 1974_spider_44  24888_spider_20  29190_spider_42 32244_spider_13  7469_bat_2
13056_spider_3  17459_spider_37 19992_spider_11 25158_spider_29  29509_spider_43 3346_bat_5       9110_spider_18
13359_spider_35 17648_spider_34 20180_spider_33 25544_spider_5   29672_spider_41 3475_spider_46   9194_bat_4
13432_spider_17 17815_spider_36 21124_spider_12 25645_spider_15  30478_spider_23 4266_spider_1    9798_spider_6
13704_spider_10 18317_spider_31 2234_spider_8   26476_spider_50  30751_spider_27 4535_spider_39   barrel_of_apples

~/Castle/Cellar
[mission 8] $ rm *_spider_*

~/Castle/Cellar
[mission 8] $ ls
14525_bat_3  18548_bat_1  3346_bat_5  7469_bat_2  9194_bat_4  barrel_of_apples

~/Castle/Cellar
[mission 8] $ gsh check

Congratulations, mission 8 has been successfully completed!

```

Missione 9 – Wildcard e file nascosti

```

~/Castle/Cellar
[mission 9] $ gsh goal

```

Mission goal

The spiders are getting clever: they found a way to hide.
Get rid of all the spiders that are hiding in the cellar without disturbing the bats.

Shell patterns

* The "*" character stands in for any sequence of characters (including an empty sequence).

? The "?" character stands in for any single character.

Remark

The wildcards "*" and "?" don't see hidden files, you need to add an explicit dot at the start of the pattern.

I ragni erano diventati file nascosti. Ho quindi utilizzato:

ls -A

rm .*_spider_*

Questa missione mi ha permesso di approfondire l'utilizzo delle wildcard e dei file nascosti.

```
~/Castle/Cellar
[mission 9] $ ls -A
.10327_spider_49 .14132_spider_12 .19400_spider_15 .24202_spider_50 3346_bat_5 7469_bat_2
.1057_spider_43 14525_bat_3 .19597_bat_5 .24472_spider_19 .3717_spider_22 .8342_spider_13
.11112_spider_23 .15184_spider_41 .19892_spider_37 .24779_spider_33 .4144_spider_28 .8472_spider_48
.11528_spider_39 .1561_spider_3 .20022_spider_16 .27759_spider_5 .45_bat_2 .8507_spider_21
.11934_spider_24 .15748_bat_1 .20642_spider_36 .28000_bat_4 .5013_spider_8 9194_bat_4
.11952_spider_31 .16405_spider_6 .21526_spider_4 .28392_spider_29 .5590_spider_11 barrel_of_apples
.12331_spider_32 .17181_spider_26 .21528_spider_38 .29672_spider_35 .5859_spider_14
.12903_spider_40 .18028_spider_46 .21647_bat_3 .30387_spider_20 .613_spider_18
.13686_spider_7 .18546_spider_47 .2175_spider_27 .31223_spider_2 .6210_spider_30
.14030_spider_45 18548_bat_1 .22405_spider_34 .32355_spider_17 .685_spider_9
.14113_spider_10 .18557_spider_1 .22644_spider_25 .32459_spider_44 .6992_spider_42

~/Castle/Cellar
[mission 9] $ rm *_spider_*
rm: cannot remove '*_spider_*': No such file or directory

~/Castle/Cellar
[mission 9] $ rm .*_spider_*

~/Castle/Cellar
[mission 9] $ ls -A
14525_bat_3 18548_bat_1 .21647_bat_3 3346_bat_5 7469_bat_2 barrel_of_apples
.15748_bat_1 .19597_bat_5 .28000_bat_4 .45_bat_2 9194_bat_4

~/Castle/Cellar
[mission 9] $ gsh check

Congratulations, mission 9 has been successfully completed!
```

Missione 10 – Copia dei file

```
~/Castle/Cellar
[mission 10] $ gsh goal

Mission goal

You have taken a fancy to the four standards in the great hall of the castle. As stealing
them would not go unnoticed, put a copy (same name, same content) of each in your chest.

Useful commands

cp FILE DIRNAME
Copy the file to the directory.
Remark: 'cp' is an abbreviation of "copy".

(*)
))
^
```

Nella Great Hall del castello erano presenti quattro standardi. Dovevo inserirne una copia nel mio Chest senza rimuovere gli originali.

Per farlo ho utilizzato:

cd

cd Castle/Great_hall

ls

cp standard_1 standard_2 standard_3 standard_4 ~/Forest/Hut/Chest

Il comando **cp** permette di **copiare** un file mantenendo l'originale nella posizione iniziale.

```

~/Castle/Cellar
[mission 10] $ cd

~
[mission 10] $ cd Great_hall
bash: cd: Great_hall: No such file or directory

~
[mission 10] $ ls
Castle/ Forest/ Garden/ Mountain/ Stall/

~
[mission 10] $ cd Castle/Great_hall

~/Castle/Great_hall
[mission 10] $ ls
1592_stag_head 227_suit_of_armour 35612_decorative_shield standard_1 standard_2 standard_3 standard_4

~/Castle/Great_hall
[mission 10] $ cp standard_1 standard_2 standard_3 standard_4 ~/Forest/Hut/Chest

~/Castle/Great_hall
[mission 10] $ gsh check

Congratulations, mission 10 has been successfully completed!

```

Missione 11 – Copia tramite pattern

```

~/Castle/Great_hall
[mission 11] $ gsh goal

()=(
(
Mission goal
=====
The tapestries in the castle's great hall are also particularly beautiful. Put a copy of
each in your chest.

Useful commands
=====
cp FILE1 FILE2 ... FILEn DIRNAME
Copy the files to the directory.
Remark: "cp" is an abbreviation of "copy".

Shell patterns
=====
*
The "*" character stands in for any sequence of characters
(including an empty sequence).

?
The "?" character stands in for any single character.

()=(
(

```

Nella Great Hall erano presenti anche degli arazzi da copiare nel Chest.

In questa missione ho combinato il comando `cp` con il comando `*` per selezionare più file contemporaneamente, evitando di doverli copiare uno alla volta.

```

~/Castle/Great_hall
[mission 11] $ ls
15161_tapestry_08 22319_stag_head 41291_tapestry_02 48334_tapestry_07 7175_decorative_shield standard_3
17799_tapestry_03 22969_tapestry_04 42524_suit_of_armour 54919_tapestry_10 standard_1 standard_4
20782_tapestry_09 3490_tapestry_06 46011_tapestry_05 6877_tapestry_01 standard_2

~/Castle/Great_hall
[mission 11] $ cp *_tapestry_* ~/Forest/Hut/Chest

~/Castle/Great_hall
[mission 11] $ gsh check

Congratulations, mission 11 has been successfully completed!

```

Missione 12 – Informazioni sui file e **cat**



Al primo piano della torre principale erano presenti diversi dipinti. L'obiettivo era individuare quello più antico.

Ho utilizzato:

`ls -l`

Il comando mi ha mostrato informazioni aggiuntive sui file, comprese le date di modifica.

Ho individuato come più antico:

`painting_JaRLesPa`

`datato 28 dicembre 1986.`

Per visualizzarne il contenuto ho utilizzato:

`cat painting_JaRLesPa`

Infine l'ho copiato nel Chest:

`cp painting_JaRLesPa ~/Forest/Hut/Chest`


```
~/Castle/Main_tower/First_floor
[mission 13] $ cal 1926

      1926
January  February  March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
1 2 1 2 3 4 5 6 1 2 3 4 5 6 1 2 3 4 5 6
3 4 5 6 7 8 9 7 8 9 10 11 12 13 7 8 9 10 11 12 13
10 11 12 13 14 15 16 14 15 16 17 18 19 20 14 15 16 17 18 19 20
17 18 19 20 21 22 23 21 22 23 24 25 26 27 21 22 23 24 25 26 27
24 25 26 27 28 29 30 28 28 29 30 31

April  May  June
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
1 2 3 1 2 3 4 5 6 7 1 2 3 4 5
4 5 6 7 8 9 10 2 3 4 5 6 7 8 6 7 8 9 10 11 12
11 12 13 14 15 16 17 9 10 11 12 13 14 15 13 14 15 16 17 18 19
18 19 20 21 22 23 24 16 17 18 19 20 21 22 20 21 22 23 24 25 26
25 26 27 28 29 30 23 24 25 26 27 28 29 27 28 29 30
31 30 31

July  August  September
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
1 2 3 1 2 3 4 5 6 7 1 2 3 4
4 5 6 7 8 9 10 8 9 10 11 12 13 14 5 6 7 8 9 10 11
11 12 13 14 15 16 17 15 16 17 18 19 20 21 12 13 14 15 16 17 18
18 19 20 21 22 23 24 22 23 24 25 26 27 28 19 20 21 22 23 24 25
25 26 27 28 29 30 31 29 30 31 26 27 28 29 30

October  November  December
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
1 2 1 2 3 4 5 6 1 2 3 4
3 4 5 6 7 8 9 7 8 9 10 11 12 13 5 6 7 8 9 10 11
10 11 12 13 14 15 16 14 15 16 17 18 19 20 12 13 14 15 16 17 18
17 18 19 20 21 22 23 21 22 23 24 25 26 27 19 20 21 22 23 24 25
24 25 26 27 28 29 30 28 29 30 26 27 28 29 30 31
31
```

```
~/Castle/Main_tower/First_floor
[mission 13] $ gsh check
What was the day of the week for the 04-20-1926?
1 : Monday
2 : Tuesday
3 : Wednesday
4 : Thursday
5 : Friday
6 : Saturday
7 : Sunday
Your answer: 2

Congratulations, mission 13 has been successfully completed!
```

Missione 14 – Creazione di un alias

```

Mission goal
-----
Checking for hidden files is taking too long!

Create an alias "la" to run the command "ls -A" in order to list all files, including
hidden ones, with only 2 letters.

Define the synonym
-----
la
-----
for the command
-----
ls -A
-----
and check that it works as expected.

How fortunate, there is a nice rock hidden just where you are.

Useful commands
-----
alias STRING='COMMAND'
Create a synonym for a string, that will stand for a command.

```

Per rendere più veloce la visualizzazione dei file nascosti ho creato un alias chiamato **la**:

```
alias la='ls -A'
```

Da quel momento è stato possibile utilizzare semplicemente:

la

per ottenere lo stesso risultato di `ls -A`.

Questa missione mi ha mostrato come gli alias possano essere utilizzati per abbreviare comandi utilizzati frequentemente.

```
~/Castle/Main_tower/First_floor
[mission 14] $ la
la: command not found

~/Castle/Main_tower/First_floor
[mission 14] $ ls -A
.nice_rock  painting_JaRlesPa  painting_KvXmfkiY  painting_uvJjgPkJ  Second_floor/

~/Castle/Main_tower/First_floor
[mission 14] $ alias la='ls -A'

~/Castle/Main_tower/First_floor
[mission 14] $ la
.nice_rock  painting_JaRlesPa  painting_KvXmfkiY  painting_uvJjgPkJ  Second_floor/

~/Castle/Main_tower/First_floor
[mission 14] $ gsh check

Congratulations, mission 14 has been successfully completed!
```

Missione 15 – Creazione e modifica di un file con Nano

```
(0) ^
=====
Mission goal
-----
Create a file named "journal.txt" in your chest and write a short message in it.
You can use this file to record your notes and solutions for the upcoming missions.

Note that "journal.txt" is a file in the "Chest" directory. You do not need to create a new
directory for this mission.

Details
-----
"nano" is a command-line text editor. You can use it whenever you need to edit a file from
the shell.

Useful commands
-----
nano FILE
Edit the file from the shell.
(If the file does not exist, it will be created.)

Keybindings are listed at the bottom of the screen (the "" symbol means "Control"). The
most important ones are:
Control-x  quit
Control-o  save
Control-w  search for a string

Remark: do not use Control-s or Control-z!
```

Nell'ultima missione svolta dovevo creare nel Chest un file chiamato:

journal.txt

Sono entrato nella directory:

cd ~/Forest/Hut/Chest

e ho aperto Nano:

nano journal.txt

All'interno del file ho inserito un breve messaggio. Ho poi salvato utilizzando **Ctrl+O**, confermato con **Invio** e chiuso Nano tramite **Ctrl+X**.

Per controllare il contenuto del file ho utilizzato:

cat journal.txt

Questa missione mi ha permesso di imparare le operazioni di base dell'editor testuale **nano**.

```
~/Castle/Main_tower/First_floor
[mission 15] $ cd ~/Forest/Hut/Chest

~/Forest/Hut/Chest
[mission 15] $ pwd
/home/kali/gameshell/World/Forest/Hut/Chest

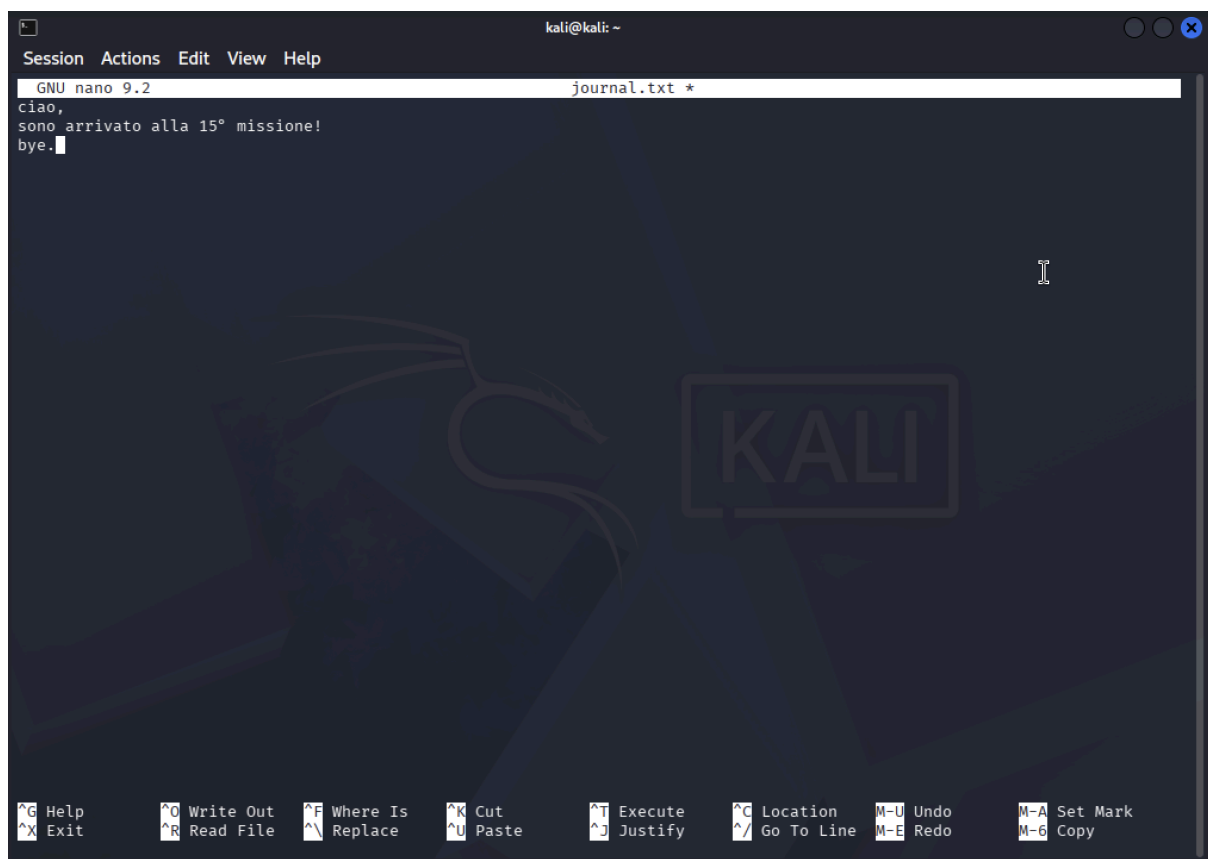
~/Forest/Hut/Chest
[mission 15] $ ls
15161_tapestry_08  22969_tapestry_04  46011_tapestry_05  6877_tapestry_01  coin_3  painting_JaRlesPa  standard_2
17799_tapestry_03  3490_tapestry_06  48334_tapestry_07  coin_1  standard_1  standard_3  standard_4
20782_tapestry_09  41291_tapestry_02  54919_tapestry_10  coin_2

~/Forest/Hut/Chest
[mission 15] $ nano journal.txt

~/Forest/Hut/Chest
[mission 15] $ ls
15161_tapestry_08  22969_tapestry_04  46011_tapestry_05  6877_tapestry_01  coin_3  journal.txt  painting_JaRlesPa  standard_1  standard_2  standard_3  standard_4
17799_tapestry_03  3490_tapestry_06  48334_tapestry_07  coin_1
20782_tapestry_09  41291_tapestry_02  54919_tapestry_10  coin_2

~/Forest/Hut/Chest
[mission 15] $ gsh check

Congratulations, mission 15 has been successfully completed!
```



4. Conclusioni

L'esercitazione con GameShell mi ha permesso di utilizzare in maniera pratica diversi comandi fondamentali di Linux. Partendo dalla semplice navigazione tra le directory, sono arrivato alla gestione di file e cartelle, alla copia e allo spostamento dei file, all'utilizzo dei caratteri jolly, alla visualizzazione dei file nascosti e alla modifica di file di testo tramite Nano.

Il metodo basato sulle missioni mi ha aiutato a comprendere non soltanto la sintassi dei singoli comandi, ma soprattutto il loro utilizzo pratico all'interno del filesystem Linux.

In totale ho proseguito l'esercitazione **fino alla missione 15**, documentando le principali operazioni effettuate come richiesto dalla traccia.

Vincenzo Lembo